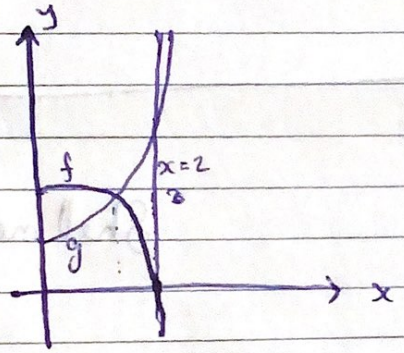
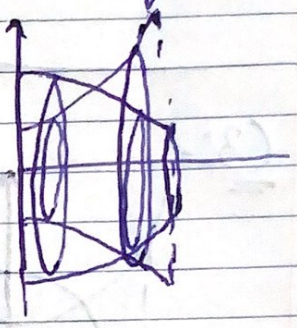
Quiz :- C2 :-

$y = 4 - x^2$, $y = x^3 + 2$ and $x = 2$, x-axis

First Quadrant :-





$4 - x^2 = x^3 + 2 \Rightarrow x^3 + x^2 + 2 - 4 = 0$

$\Rightarrow x^3 + x^2 - 2 = 0 \Rightarrow x^2(x+1) - 2 = 0$

$\Rightarrow x^2(x+1) = 2 \Rightarrow x^2 = 2 , (x+1) = 2$

$\boxed{x = \pm\sqrt{2}}$, $\boxed{x = 1}$

$(x^2-2)(x+1)$

$x^3 - 2x + x^2 - 2$

$V = \pi \int_0^1 \left[(4 - x^2)^2 - (x^3 + 2)^2 \right] dx$

$+ \pi \int_1^2 \left[(x^3+2)^2 - (4 - x^2)^2 \right] dx$

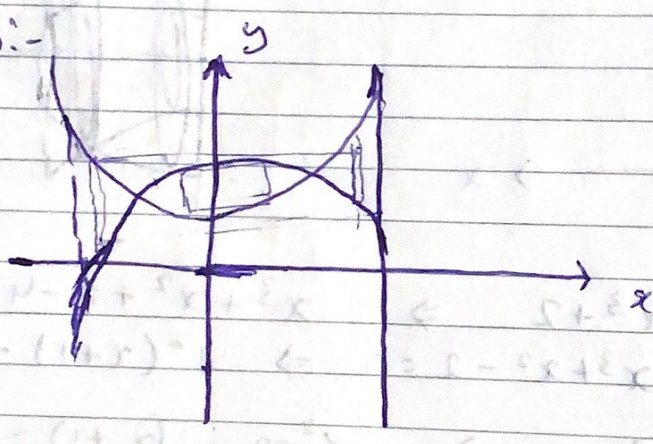
$V = \pi \int_0^1 \left[(16 - 8x^2 + x^4) - [x^6 + 4x^3 + 4) \right] dx$

$+ \pi \int_1^2 \left[(x^6 + 4x^3 + 4) - (16 - 8x^2 + x^4) \right] dx$

$$V = \pi \int_0^1 (12 - x^6 + x^4 - 4x^3 - 8x^2)\,dx$$

$$+ \pi \int_1^2 (x^6 - x^4 + 4x^3 + 8x^2 - 12)\,dx$$

(3:-



Shell method :-

$$V = \int_a^b 2\pi (R)(H)\,dx$$

$$V = \int_0^2 2\pi (x)\left((4 - x^2) - (x^3 + 2)\right)dx$$